

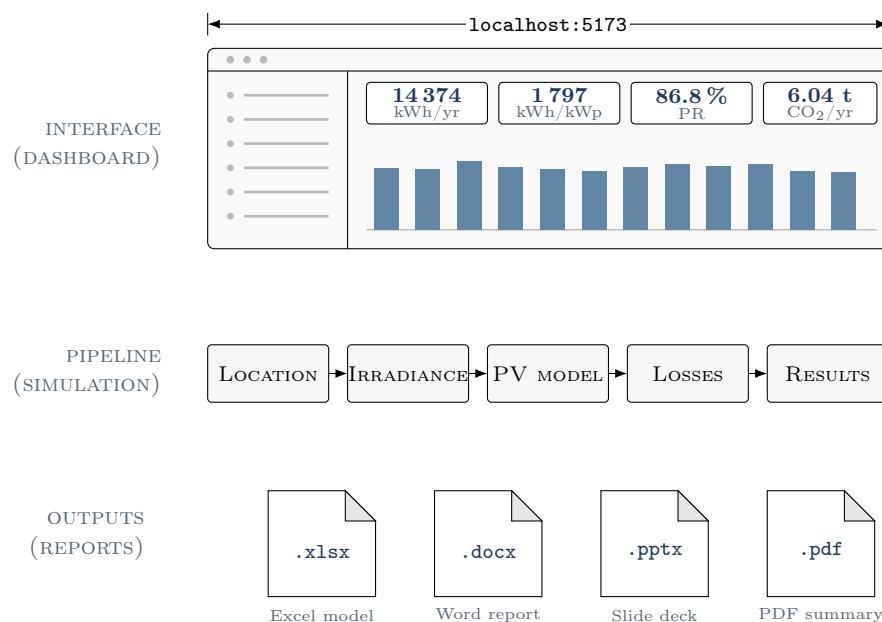
TECHNICAL ENGINEERING REPORT

Renewable Energy · Software Systems · Financial Modeling

SolarPV Studio

A Local Web Application for Solar Photovoltaic System Simulation

Systems Architecture, Solar Physics, Energy Yield, Financial Modeling, and Automated Reporting



General arrangement of the application (not to scale). The dashboard interface, the physics simulation pipeline, and the four generated deliverables.

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1 Introduction

SolarPV Studio is a private, fully local web application for the design and simulation of solar photovoltaic (PV) systems, inspired by the workflow of professional tools such as PVsyst. The application runs entirely on the engineer’s machine (`localhost:5173`); no project data leaves the computer. It covers the complete study cycle:

1. site selection and solar-resource assessment,
2. PV array and inverter configuration with loss budgeting,
3. physics-based annual energy-yield simulation,
4. financial analysis over a 25-year horizon, and
5. automated generation of four professional deliverables (Excel financial model, Word report, PowerPoint deck, PDF summary).

This report documents the software architecture (Section 2), the solar physics engine (Section 3), a verified case study in Dubai (Section 4), the loss and financial models (Sections 5–6), and the report-generation engine (Section 7).

2 System Architecture

The application is a single-page React application served by Vite. All computation is client-side JavaScript; simulation results persist in browser local storage, so a project survives restarts without any database.

Layer	Technology	Role
Interface	React 19 + Tailwind CSS	Six-page dark-theme UI
Charts	Recharts	Interactive result charts
Physics	Custom JavaScript modules	Solar position, irradiance, yield
Persistence	Browser <code>localStorage</code>	Project auto-save
Reports	ExcelJS, docx-js, pptxgenjs, pdf-lib	Four deliverables
AI chat	Claude API (streaming)	Project-aware assistant

Table 1: Technology stack.

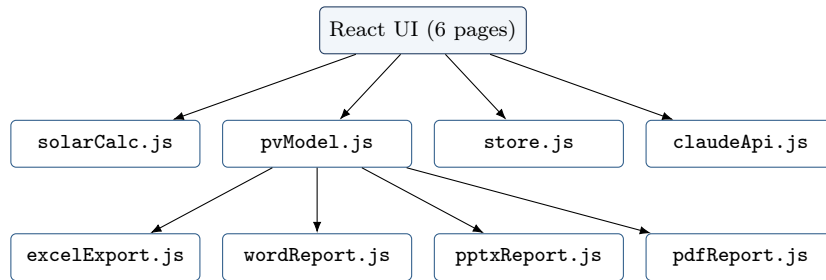


Figure 1: Module architecture. The UI drives the physics core; the physics core feeds all four report generators.

3 Solar Physics Engine

3.1 Solar position

For day of year N and phase angle $B = \frac{360}{365}(N - 81)$, the solar declination and the equation of time are

$$\delta = 23.45^\circ \sin B, \quad (1)$$

$$\text{EoT} = 9.87 \sin 2B - 7.53 \cos B - 1.5 \sin B \quad [\text{min}]. \quad (2)$$

With local solar noon $t_{sn} = 12 - \text{EoT}/60 - (\lambda \bmod 15)/15$ the hour angle is $\omega = 15^\circ(t - t_{sn})$, and the solar elevation α at latitude φ follows from

$$\sin \alpha = \sin \varphi \sin \delta + \cos \varphi \cos \delta \cos \omega. \quad (3)$$

3.2 Clear-sky irradiance

The relative air mass uses the Kasten–Young formulation,

$$\text{AM} = [\sin \alpha + 0.50572 (\alpha + 6.07995)^{-1.6364}]^{-1}, \quad (4)$$

and with the extraterrestrial irradiance $E_0 = 1367(1 + 0.033 \cos \frac{360N}{365})$ W/m² and site altitude h , the beam, diffuse and global components are

$$\text{DNI} = E_0 e^{-0.090 \text{AM}^{0.84}} e^{-h/8500}, \quad (5)$$

$$\text{DHI} = 0.0734 E_0 (\cos \alpha)^{1.01} e^{-h/8500}, \quad (6)$$

$$\text{GHI} = \text{DNI} \sin \alpha + \text{DHI}. \quad (7)$$

3.3 Transposition to the array plane

For tilt β and azimuth γ (module) and γ_s (sun), the incidence factor is

$$\cos \theta = \cos \alpha \cos(\gamma_s - \gamma) \sin \beta + \sin \alpha \cos \beta, \quad (8)$$

and the plane-of-array irradiance with ground albedo $\rho = 0.2$ is

$$G_t = \text{DNI} \max(0, \cos \theta) + \text{DHI} \frac{1 + \cos \beta}{2} + \text{GHI} \rho \frac{1 - \cos \beta}{2}. \quad (9)$$

Clear-sky values are scaled by a monthly clearness index $K_t(m)$ calibrated per climate zone (0.60–0.68 for hot arid climates) to account for cloud cover.

3.4 Thermal model and energy yield

Cell temperature follows the NOCT model, $T_{cell} = T_{amb} + \frac{\text{NOCT}-20}{800} G$, giving a thermal derating factor $f_T = 1 + \frac{\gamma_P}{100}(T_{cell} - 25)$ for a module power coefficient γ_P [%/K]. Monthly energy is then

$$E_m = \bar{H}_{t,d} K_t P_{STC} f_{loss} f_T N_d, \quad f_{loss} = (1 - l_s)(1 - l_w)(1 - l_m)(1 - l_{sh}) a \eta_{inv}, \quad (10)$$

where l_s, l_w, l_m, l_{sh} are soiling, wiring, mismatch and shading fractions, a availability, and η_{inv} the inverter efficiency. The simulation runs at 30-minute resolution across a representative day per month.

4 Case Study: 8 kWp Rooftop in Riyadh

Parameter	Value	Parameter	Value
Location	Riyadh, KSA (24.71355°N, 46.67530°E)	Modules	20 × JinkoSolar 400 Wp
Altitude	5 m	Array power	8.00 kWp
Tilt	25°	Inverter	SMA Sunny Boy 5.0 (97 %)
Azimuth	180° (south)	Loss inputs	soiling 2 %, wiring 2 %, mismatch 1 %

Table 2: Simulation configuration.

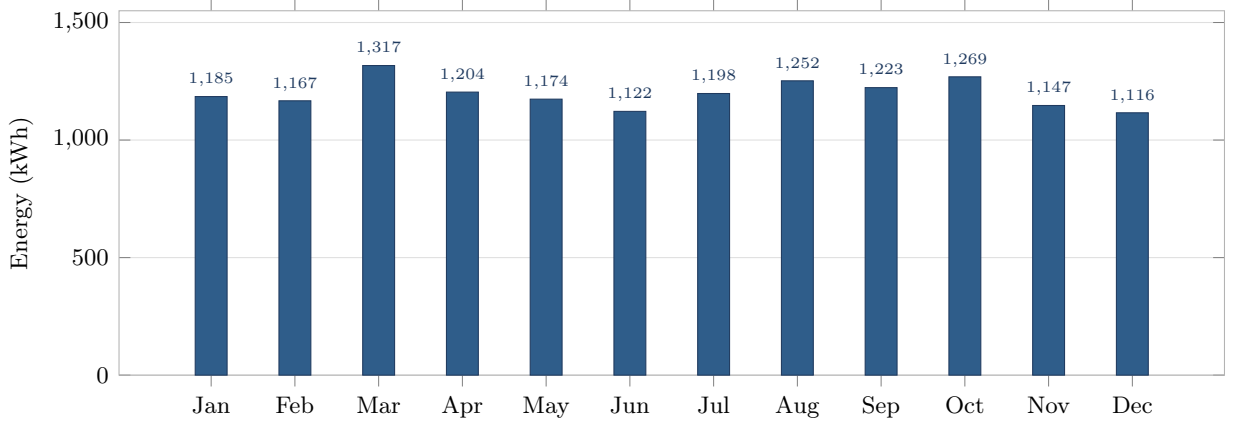


Figure 2: Simulated monthly AC energy production (total 14 374 kWh/yr).

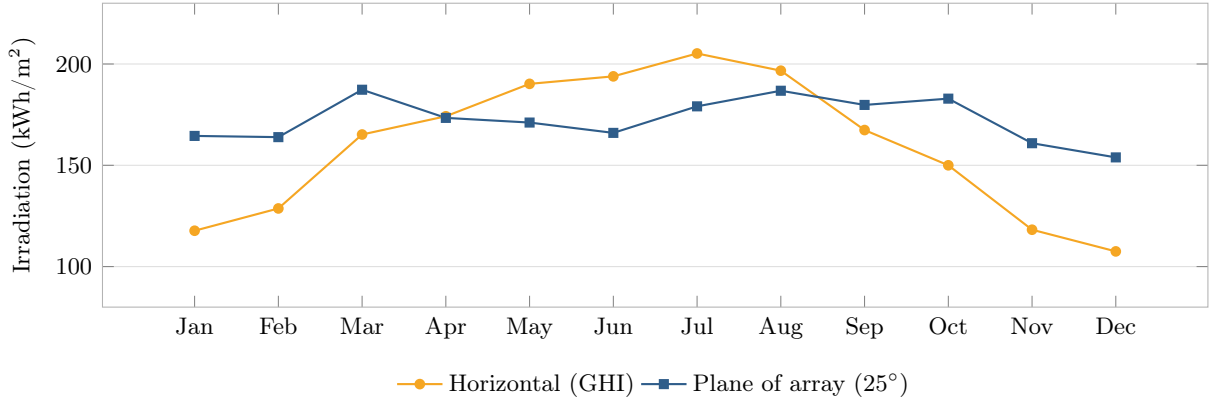


Figure 3: Monthly irradiation: the 25° tilt flattens the seasonal profile, trading summer excess for winter gain (+8.1 % annually).

Month	GHI (kWh/m ²)	Tilted (kWh/m ²)	Energy (kWh)	Specific (kWh/kWp)	T_{amb} (°C)
Jan	117.7	164.5	1 185	148	15
Feb	128.7	163.9	1 167	146	17
Mar	165.2	187.3	1 317	165	20
Apr	174.2	173.4	1 204	151	24
May	190.2	171.1	1 174	147	28
Jun	193.9	166.0	1 122	140	32
Jul	205.2	179.1	1 198	150	34
Aug	196.7	186.8	1 252	156	33
Sep	167.4	179.8	1 223	153	29
Oct	150.0	182.9	1 269	159	24
Nov	118.2	160.9	1 147	143	18
Dec	107.5	153.9	1 116	140	14
Year	1 915	2 070	14 374	1 797	–

Table 3: Monthly simulation results.

Annual performance indicator	Value
AC energy	14 374 kWh/yr
Specific yield	1 797 kWh/kWp
Performance ratio	86.8 %
Capacity factor	20.5 %
CO ₂ avoided (0.42 kg/kWh)	6.04 t/yr

Table 4: Key performance indicators.

5 Loss Analysis

Total system losses amount to 1918 kWh/yr (11.8% of the ideal yield). Temperature derating dominates, as expected for a hot arid site.

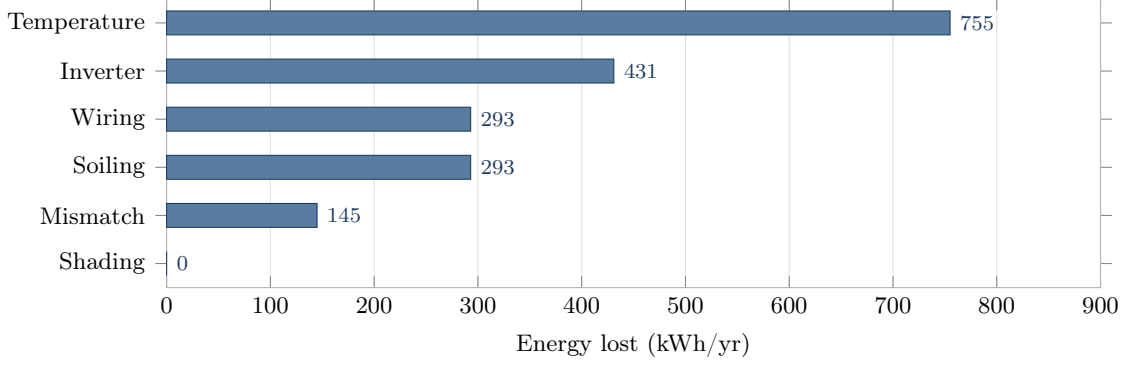


Figure 4: Annual loss breakdown by mechanism.

6 Financial Model

Year- y revenue combines linear module degradation d with tariff escalation e on tariff p :

$$R_y = E_1 (1 - d y) p (1 + e)^{y-1}, \quad d = 0.5 \%/\text{yr}, \quad e = 2 \%/\text{yr}, \quad p = 0.12 \$/\text{kWh}. \quad (11)$$

With capital cost $C_0 = \$7\,200$ (900 \$/kWp) the project value follows

$$\text{NPV} = \sum_{y=1}^{25} \frac{R_y}{(1 + r)^y} - C_0, \quad \text{LCOE} = \frac{C_0}{\sum_{y=1}^{25} E_y}, \quad r = 5 \%. \quad (12)$$

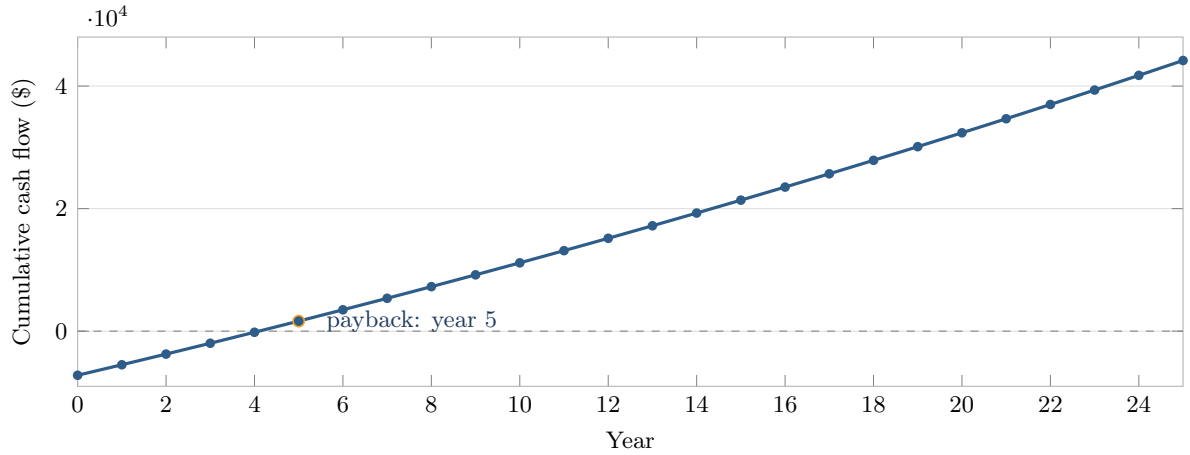


Figure 5: 25-year cumulative cash flow; break-even occurs in year 5.

Financial indicator	Value
System cost	\$7 200
Year-1 revenue	\$1 725
Simple payback	5 years
Net present value (25 yr)	\$20 736
Internal rate of return	10.9 %
Levelized cost of energy	2.14 ¢/kWh

Table 5: Financial results (assumptions editable in the Excel model).

7 Report Generation Engine

A dedicated Report Center generates four deliverables locally, each through the reference library of the corresponding document skill:

Deliverable	Format	Library	Highlights
Financial model	.xlsx	ExcelJS	126 live formulas, named inputs, conditional formatting
Engineering report	.docx	docx-js	7 sections, styled tables
Proposal deck	.pptx	pptxgenjs	6 slides, native charts
Summary	.pdf	pdf-lib	2 pages, KPI boxes, tables

Table 6: Report Center deliverables.

The Excel workbook is a live model: tariff, escalation, discount rate, degradation and cost per kWp are editable named cells; payback, NPV, IRR and the 25-year cash flow recalculate inside Excel. The workbook was verified with a LibreOffice recalculation pass: **zero formula errors across all 126 formulas**.

8 AI Assistant

An integrated chat assistant (Claude API) receives the full project state – location, configuration, simulation results and loss budget – as a system context, and answers optimization questions such as tilt trade-offs, loss prioritization and tariff sensitivity. Without an API key the assistant falls back to a deterministic demo mode; responses stream token-by-token in both modes.

9 Conclusions

SolarPV Studio demonstrates that a professional PV study workflow – resource assessment, physics simulation, financial modeling and client-ready reporting – can run entirely in a local browser with no cloud dependency. For the Dubai reference case the tool predicts **14 374 kWh/yr** (1 797 kWh/kWp, PR 86.8%) with break-even in year 5 and a 25-year NPV of \$20 736. Planned work includes horizon-line shading, bifacial gain modeling, and TMY weather-file import.

A Equipment Database (excerpt)

Module	Technology	P_{max} (Wp)	η (%)	γ_P (%/K)
JinkoSolar Eagle 78TR	Mono PERC	400	20.4	−0.35
LONGi Hi-MO 6	Mono PERC	540	21.3	−0.34
SunPower Maxeon 3	Mono IBC	420	22.7	−0.27
Trina Vertex S+	TOPCon	450	22.5	−0.30
JinkoSolar Tiger Neo	TOPCon	575	22.3	−0.29
Canadian Solar BiHiKu7	Bifacial	600	21.2	−0.34

Table 7: PV module database (excerpt).